

2/3 UNIT MATHEMATICS FORM VI

Time allowed: 3 hours

Exam date: 4th August, 1997

Instructions:

- All questions may be attempted.
- All questions are of equal value.
- Part marks are shown in boxes in the left margin.
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Collection:

- Each question will be collected separately.
- Start each question in a new answer booklet.
- If you use a second booklet for a question, place it inside the first. Don't staple.
- Write your candidate number on each answer booklet.

QUESTION THREE (Start a new answer booklet)

Marks

3 (a) Find $\frac{dy}{dx}$ given:

(i) $y = \tan(3x + 5),$

(ii) $y = \log_e(2x + 1),$

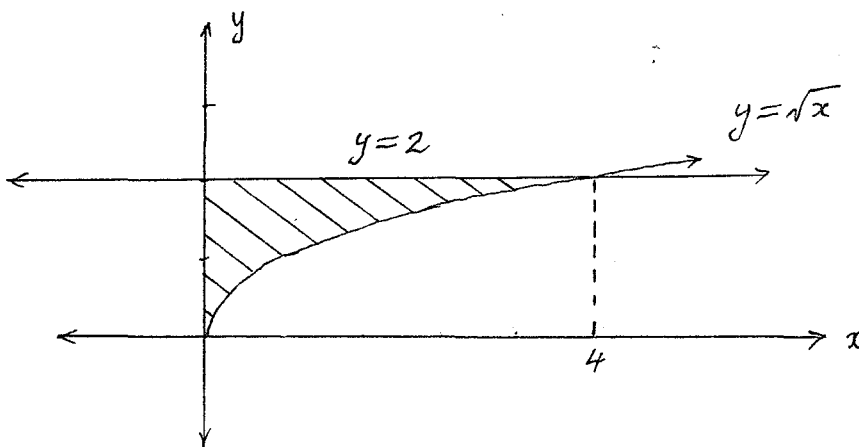
(iii) $y = \frac{e^x}{x}.$

3 (b) (i) Find $\int \cos 2x \, dx.$

(ii) Find $\int_0^1 \frac{2}{x+1} \, dx.$

3 (c) Use the relationship $\tan^2 x + 1 = \sec^2 x$ to evaluate $\int_0^{\frac{\pi}{4}} \tan^2 x \, dx.$

3 (d)



The diagram shows the area bounded by the y axis, the curve $y = \sqrt{x}$ and the line $y = 2$. Find the area of the shaded region.

QUESTION FIVE (Start a new answer booklet)

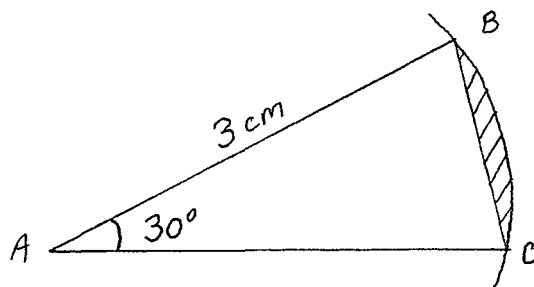
Marks

- 4** (a) Consider the parabola $4y = x^2 - 4x$.
- (i) Show algebraically how the parabola can be expressed in the form $(x - 2)^2 = 4(y + 1)$.
 - (ii) Write down the co-ordinates of the focus.
 - (iii) Find the equation of the directrix.
- 4** (b) The n th term of an arithmetic sequence is given by $U_n = 2n - 11$.
- (i) Find the first term and the common difference of the sequence.
 - (ii) Calculate the sum of the series to the fifteenth term.
- 4** (c) If α and β are the roots of the quadratic equation $x^2 - 2x - 1 = 0$ find the value of:
- (i) $\frac{1}{\alpha} + \frac{1}{\beta}$,
 - (ii) $\alpha^2 + \beta^2$.

QUESTION SIX (Start a new answer booklet)

Marks

- 8** (a) Consider the curve $y = 2x^3 + 3x^2 - 12x + 2$.
- (i) Find all stationary points and determine their nature.
 - (ii) Find any points of inflexion.
 - (iii) Sketch the curve for $-3 \leq x \leq 3$, showing the y -intercept.
 - (iv) For what values of x is the curve increasing but concave down.
- 4** (b)



In the diagram above $\angle BAC = 30^\circ$ and a circular arc of radius 3 cm and centre A is constructed from B to point C .

- (i) Find the area of $\triangle ABC$.
- (ii) Calculate the exact area of the shaded segment.

QUESTION NINE (Start a new answer booklet)

Marks

- 4** (a) A particle moves with an acceleration given by $\ddot{x} = \sqrt{t} - \frac{1}{\sqrt{t}}$. Initially the velocity is $\frac{4}{3}$ m/s and the displacement is $\frac{4}{3}$ m.
- (i) Express the velocity $\dot{x}$ in terms of t .
- (ii) Find the displacement x when $t = 1$ sec.

- 3** (b) It costs a manufacturer $\$c$ to make and distribute a calculator. The item sells at $\$x$ each, and the total number sold is given by:

$$n = \frac{a}{x - c} + b(100 - x),$$

where a, b and c are positive constants. Find the selling price that will bring the maximum profit.

- 5** (c) A man borrows $\$40\,000$ from a building society at an interest rate 6% per annum compounded monthly. The loan will be repaid over ten years by equal monthly instalments of $\$Q$. Let $R = 1 + \frac{0.06}{12}$.

- (i) Show that the total amount owing A after n months is given by:

$$A = 40\,000R^n - Q(1 + R + R^2 + \dots + R^{n-1}).$$

- (ii) From this expression, calculate the monthly repayments for the loan to be repaid after ten years.

(d)

$$A = 8 - \int_0^4 x^{\frac{1}{2}} dx \quad \checkmark$$

$$= 8 - \left[\frac{2x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^4 \quad \checkmark$$

$$= 8 - \left[\frac{16}{3} - 0 \right]$$

$$= 2\frac{2}{3} \text{ sq units. } \checkmark$$

OR //

$$\text{Area} = \int_0^2 y^2 dy$$

$$= \left[\frac{1}{3} y^3 \right]_0^2$$

(12)

$$= 2\frac{2}{3} \text{ sq units.}$$

QUESTION 4

(a)

(i) $\theta = 27^\circ 45'$

$$\overline{CL}^2 = 120^2 + 260^2 - 2 \times 120 \times 260 \times \cos 27^\circ 45' \quad \checkmark$$

$$CL = 164 \text{ (nearest km)} \quad \checkmark$$

(ii) let $\alpha = \angle CLA$

$$\frac{\sin \alpha}{120} = \frac{\sin 27^\circ 45'}{164} \quad \checkmark$$

$$\sin \alpha = \frac{120 \sin 27^\circ 45'}{164}$$

$$\alpha = 19^\circ 56' \quad \checkmark$$

$$\text{Bearing} = 270^\circ + 19^\circ 56'$$

$$= 289^\circ 56' \text{ (nearest minute)} \quad \checkmark$$

(b)

(i) In $\triangle NMO$ and $\triangle MLP$

$\angle LMP = \angle MON$ (alternate angles of

$\angle LPM = \angle NMP$ (parallel lines $\checkmark$)

$\therefore \triangle MNO \parallel \triangle PLM$ (corresponding angles are equal). $\checkmark$

(ii) $\frac{x+2}{x} = \frac{12}{9} \quad \checkmark$ (Corresponding sides are in the same ratio)

$$12x = 9x + 18$$

$$3x = 18$$

$$x = 6 \text{ units. } \checkmark$$

(c) $2x - 3y + 1 + k(x + y - 3) = 0$

$$-2 - 6 + 1 + k(-1 + 2 - 3) = 0 \quad \checkmark$$

$$-7 - 2k = 0$$

$$k = -\frac{7}{2} \quad \checkmark$$

The equation of line through the pt of intersection:

$$2x - 3y + 1 - \frac{7}{2}(x + y - 3) = 0$$

$$4x - 6y + 2 - 7x - 7y + 21 = 0$$

$$-3x - 13y + 23 = 0$$

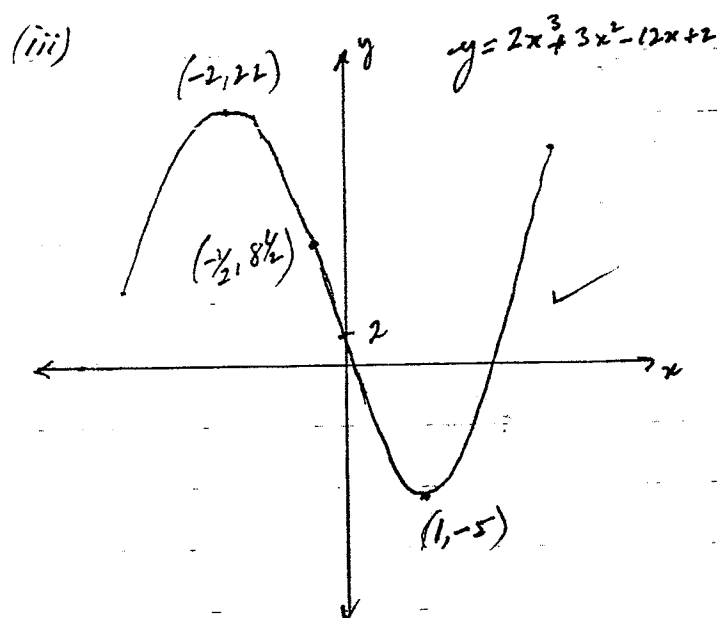
$$3x + 13y - 23 = 0 \quad \checkmark$$

(12)

Test for a change in concavity.

x	-1	$-\frac{1}{2}$	1
$f''(x)$	-	0	+

∴ $f(-\frac{1}{2} + \epsilon) > 0$
 $f(-\frac{1}{2} - \epsilon) < 0$ ✓
 point of inflexion. $(-\frac{1}{2}, 8\frac{1}{2})$



(iv) Concave down.

$$\{x: x < -\frac{1}{2}\}$$

(b)

(i) Area $\triangle ABC = \frac{1}{2} \times 3 \times 3 \sin 30^\circ$
 $= \frac{9}{4} \text{ cm}^2$ ✓

(ii)

Segment = Area - Area $\triangle ABC$
 Area Sector

$$= \frac{1}{2} r^2 \theta - \frac{9}{4}$$

$$= \frac{1}{2} \times 9 \times \frac{\pi}{6} - \frac{9}{4}$$

(12)

$$A = \frac{3}{4} (\pi - 3) \text{ cm}^2$$

QUESTION 7

(a)

(i) $x^2 + 6x + k + 8$

$$\Delta = b^2 - 4ac \quad \checkmark$$

$$\Delta = 36 - 4(k + 8)$$

$$\Delta = 4 - 4k \quad \checkmark$$

(ii) $y = 4x + k$ is a tangent to the parabola $y = -8 - 2x - x^2$ if there is only one point of intersection.

$$4x + k = -8 - 2x - x^2 \quad \checkmark$$

$$x^2 + 6x + k + 8 = 0$$

one root.

$$\Delta = 0$$

$$4 - 4k = 0$$

$$k = 1 \quad \checkmark$$

(b) 27, 18, 12, ...

(i) $a = 27$

$$r = \frac{2}{3}$$

$$S_n = \frac{a(1-r^n)}{(1-r)}$$

$$S_8 = \frac{27(1-(\frac{2}{3})^8)}{\frac{1}{3}} \quad \checkmark$$

$$= 81(1 - \frac{256}{38}) \quad \checkmark$$

$$= 77 \frac{68}{81} \text{ cm.}$$

(ii) $S_\infty = \frac{a}{1-r} \quad \checkmark$

$$= \frac{27}{\frac{1}{3}}$$

$$= 81 \text{ cm.} \quad \checkmark$$

QUESTION 9

$$?) \ddot{x} = t^{1/2} - t^{-1/2}$$

$$(i) \ddot{x} = \frac{2}{3} t^{3/2} - 2t^{1/2} + c_1 \checkmark$$

$$\text{When } t=0, \ddot{x} = \frac{4}{3}$$

$$\frac{4}{3} = c_1$$

$$\ddot{x} = \frac{2}{3} t^{3/2} - 2t^{1/2} + \frac{4}{3} \checkmark$$

$$(ii) x = \frac{4}{15} t^{5/2} - \frac{4}{3} t^{3/2} + \frac{4}{3} t + c_2$$

$$x = \frac{4}{3} \text{ when } t=0$$

$$c_2 = \frac{4}{3}$$

$$x = \frac{4}{15} t^{5/2} - \frac{4}{3} t^{3/2} + \frac{4}{3} t + \frac{4}{3} \checkmark$$

$$\text{When } t=1,$$

$$x = \frac{4}{15} - \frac{4}{3} + \frac{4}{3} + \frac{4}{3}$$

$$x = 1\frac{3}{5} m \checkmark$$

b)

$$\text{Profit/calculator} = x - c$$

$$\text{Total Profit} = (x - c)n$$

$$P = (x - c) \left\{ \frac{a}{(x - c)} + b(100 - x) \right\}$$

$$P = a + b(100 - x)(x - c) \checkmark$$

$$P = a + b(100x - 100c - x^2 + xc)$$

$$P = a + 100bx - 100bc - bx^2 + bcx$$

$$\frac{dP}{dx} = 100b - 2bx + bc$$

$$100b - 2bx + bc = 0$$

$$2bx = 100b + bc$$

$$2x = 100 + c$$

$$\checkmark x = \frac{100 + c}{2} //$$

$$\frac{d^2P}{dx^2} = -2b$$

$$\frac{d^2P}{dx^2} = -2b < 0 \text{ local max.}$$

(c)

(i) A_n - the amount owing after n months.

$$A_1 = 40000R - Q \checkmark$$

$$A_2 = (40000R - Q)R - Q$$

$$= 40000R^2 - QR - Q$$

$$A_3 = (40000R^2 - QR - Q)R - Q$$

$$= 40000R^3 - QR^2 - QR - Q$$

$$= 40000R^3 - Q(1 + R + R^2) \checkmark$$

$$A_n = 40000R^n - Q(1 + R + R^2 + \dots + R^{n-1}) \checkmark$$

$$(ii) A_{120} = 0$$

$$Q = \frac{40000R^{120}}{(1 + R + R^2 + \dots + R^{n-1})} \text{ G.P.}$$

$$Q = \frac{40000R^{120}(R - 1)}{R^n - 1} \checkmark$$

$$Q = \frac{40000(1.005)^{120} \times 0.005}{(1.005)^{120} - 1}$$

$$Q = \$444.08 / \text{month} \checkmark$$

(nearest cent)

(12)